

Report preview verification

REF
CAT-PREVIEW

ASSESSOR
David Kemp

ASSESSED
2026-05-31

NEXT REVIEW
2027-05-31

Generated May 31, 2026 with CAT — COSHH Assessment Tool

Research Lab

01 OVERVIEW

BUSINESS UNIT Research Lab	LOCATION Chemistry prep room	RISK ASSESSOR David Kemp	SOP REF NUMBER(S) SOP-PREVIEW
DATE OF ASSESSMENT 2026-05-31	DATE OF REVIEW 2027-05-31	PERSONS AT RISK Staff, Students	
ACTIVITY OUTLINE Small-scale preparation and disposal of mixed laboratory chemicals.			

02 PROCESS STEPS & CHEMICALS

STEP	ACTIVITY	CHEMICALS	GHS SUMMARY	ENGINEERING	PPE	OTHER CONTROLS
Step 1	Prepare and transfer chemicals Dispense small quantities in a fume hood.	5		Fume hood	Gloves, Goggles, Lab coat	Use secondary containment.

03 TASK HAZARDS

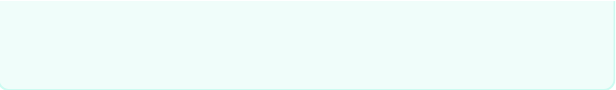
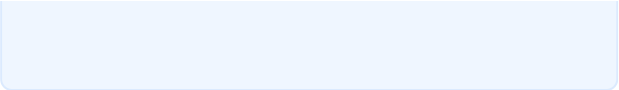
HAZARD	HOW HARM OCCURS	BEFORE CONTROLS	CONTROLS IN PLACE	AFTER CONTROLS	FURTHER ACTION
Glassware breakage during transfer	Cuts and secondary chemical release.	L3 x S3 = 9	Use trays and clear bench space.	L1 x S2 = 2	Brief staff before task.

04 CONTROLS

Elimination / substitution / reduction
Elimination: Remove any chemical not required by the method.
Substitution: Reviewed; no suitable lower hazard substitute identified.
Reduction: Use minimum working quantities.

Engineering / administration
Engineering: Use fume hood for dispensing and transfer.
Admin: Only trained authorised personnel may complete the task. Check SDS before work.

PPE / monitoring / health
PPE: Chemical resistant gloves, goggles and lab coat.
Standard: Confirm glove compatibility against SDS.
Air monitoring: Review WEL and task conditions.



Health surveillance: Review SDS and Occupational Health triggers.

05 STORAGE

STORAGE

Use assigned storage groups below;
keep minimum stock.

INCOMPATIBLE SUBSTANCES

Segregate acids, bases, oxidisers,
flammables and toxics unless SDS
confirms compatibility.

Per-chemical storage assignments from the Storage page. Confirm each row against SDS sections 7 and 10 and local storage rules.

Acetone

CONFIRMED

Flammable Cabinet

Hazards: Flammable

Segregation: Oxidizers, strong acids/bases, inorganic toxins and reactive materials unless SDS confirms compatibility.

Hydrochloric acid

CONFIRMED

Corrosives Cabinet (Acids)

Hazards: Corrosive acid

Segregation: Bases, cyanides, sulphides, flammables, organic acids and reactive materials unless SDS confirms compatibility.

Sodium hydroxide

CONFIRMED

Corrosives Cabinet (Bases)

Hazards: Corrosive base

Segregation: Acids, flammables, organic toxins and reactive materials unless SDS confirms compatibility.

Hydrogen peroxide 30%

CONFIRMED

Oxidizers Cabinet

Hazards: Corrosive, Oxidising

Segregation: Flammables, organic materials, organic acids, organic toxins and reactive materials unless SDS confirms compatibility.

Phenol

CONFIRMED

Toxins Cabinet - organic

Hazards: Corrosive, Toxic

Segregation: Acids, bases, oxidizers, inorganic toxins and reactive materials unless SDS confirms compatibility.

06 EMERGENCY RESPONSE AND WASTE

SPILLS	Use SDS Section 6 and local spill procedure.
FIRST AID	Use SDS Section 4 as definitive first-aid instruction.
FIRE	Use SDS Section 5; provide SDS to responders.
WASTE HANDLING	Collect compatible waste streams in labelled closed containers; confirm segregation via SDS.

OTHER

Confirm eyewash, shower, spill kit and SDS access before starting.

07 BRIEFING & SIGN-OFF

NAME

DATE

SIGNATURE

Test user

2026-05-31

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A1 APPENDIX A — CHEMICAL DETAIL

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	EXPOSURE	GHS
1	Acetone CAS 67-64-1	liquid 250 mL	H225; H319; H336	Check EH40/SDS / — Manual-EH40	Up to 30 minutes Weekly inhalation, skin, eye	 
2	Hydrochloric acid CAS 7647-01-0	liquid 100 mL	H290; H314; H335	Check EH40/SDS / — Manual-EH40	Up to 30 minutes Weekly inhalation, skin, eye	 
3	Sodium hydroxide CAS 1310-73-2	solid 100 g	H290; H314	Check EH40/SDS / — Manual-EH40	Up to 30 minutes Weekly inhalation, skin, eye	
4	Hydrogen peroxide 30% CAS 7722-84-1	liquid 50 mL	H272; H302; H318	Check EH40/SDS / — Manual-EH40	Up to 30 minutes Weekly inhalation, skin, eye	  
5	Phenol CAS 108-95-2	solid 25 g	H301; H311; H314; H331; H341; H373	Check EH40/SDS / — Manual-EH40	Up to 30 minutes Weekly inhalation, skin, eye	  

B1 APPENDIX B — COSHH ESSENTIALS EXPLANATION & SCREENING

COSHH Essentials is an HSE control-banding screen. CAT presents it as substance-level screening: it estimates a control approach for each substance from health hazard group, quantity scale, and volatility or dustiness. It is not a legal approval by itself: a competent assessor must confirm the output against SDS information, exposure route, quantity, duration, WELs and local conditions.

Approaches present

Approach 1, Approach 2, Approach 4

Highest screening output

Approach 4 — Specialist advice required. Driven by Phenol.

Assessor confirmation

Verify each chemical against SDS, task, route, quantity, duration, WELs and local conditions.

APPROACH 1	General ventilation and good practice may be sufficient only when confirmed by a competent assessor for the actual task.
APPROACH 2	Engineering control is indicated, normally LEV or equivalent capture/control.
APPROACH 3	Containment or enclosure is indicated where small breaches may occur.
APPROACH 4	Specialist advice is required; do not rely on the banded screening alone.

SUBSTANCE	GROUP	H-CODES	SCALE	BAND	EP	APPROACH
Acetone	A	—	small	high (volatility)	EP2	Approach 1 — General ventilation
Hydrochloric acid	C	H314, H335	small	medium (volatility)	EP2	Approach 2 — Engineering controls (LEV)
Sodium hydroxide	C	H314	small	medium (dustiness)	EP1	Approach 1 — General ventilation
Hydrogen peroxide 30%	C	H318	small	low (volatility)	EP1	Approach 1 — General ventilation
Phenol	E	H341	small	medium (dustiness)	EP1	Approach 4 — Specialist advice required